



AI WITHOUT FEAR: SECURE CODE EXECUTION FOR LLMS IN AZURE CONTAINER APPS

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EXAMPLE: CHATGPT CODE EXECUTION

Plot the formula $y=mx+b$ with m being 4 and b being 2

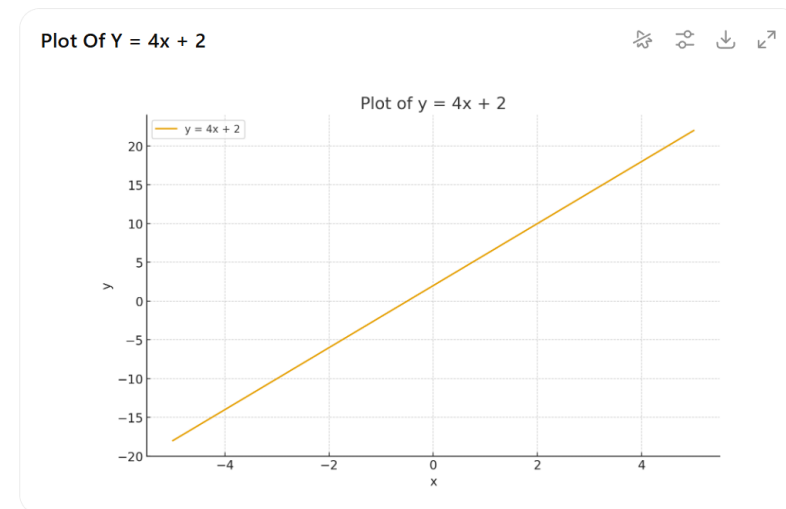
```
Analysis
python Always show details Copy code

import matplotlib.pyplot as plt
import numpy as np

# Parameters
m = 4
b = 2

# Generate x values
x = np.linspace(-5, 5, 100)
y = m * x + b

# Plot
plt.plot(x, y, label=f"y = {m}x + {b}")
plt.title("Plot of y = 4x + 2")
plt.xlabel("x")
plt.ylabel("y")
plt.grid(True)
plt.legend()
plt.show()
```



Here's the plot of the line $y = 4x + 2$. The line has a slope of 4 (steep rise) and a y-intercept at 2. [↗](#)

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python

Always show details



Copy code

```
from reportlab.platypus import SimpleDocTemplate, Image, Paragraph
from reportlab.lib.pagesizes import letter
from reportlab.lib.styles import getSampleStyleSheet
```

```
# Save the plot as an image first
```

```
image_path = "/mnt/data/line_plot_y_4x_2.png"
```

```
plt.plot(x, y, label=f"y = {m}x + {b}")
```

```
plt.title("Plot of y = 4x + 2")
```

```
plt.xlabel("x")
```

```
plt.ylabel("y")
```

```
plt.grid(True)
```

```
plt.legend()
```

```
plt.savefig(image_path, bbox_inches="tight")
```

```
plt.close()
```

```
# Create PDF
```

```
pdf_path = "/mnt/data/line_plot_y_4x_2.pdf"
```

```
doc = SimpleDocTemplate(pdf_path, pagesize=letter)
```

```
styles = getSampleStyleSheet()
```

```
story = []
```

```
# Add title and image
```

```
story.append(Paragraph("Plot of y = 4x + 2", styles["Title"]))
```

```
story.append(Spacer(1, 12))
```

```
story.append(Image(image_path, width=400, height=300))
```

```
story.append(Spacer(1, 12))
```

```
story.append(Paragraph("Equation: y = 4x + 2", styles["Normal"]))
```

```
story.append(Paragraph("This line has a slope (m) of 4 and a y-intercept of 2"))
```

```
doc.build(story)
```

```
pdf_path
```

Result

```
'/mnt/data/line_plot_y_4x_2.pdf'
```

EXAMPLE: CHATGPT FILE HANDLING

[Generate a PDF of this graph](#)

Here's your PDF of the graph: [Download line_plot_y_4x_2.pdf](#)



Demo

- ChatGPT code execution

So, you want to let an AI write and run code?

The challenge

- LLMs often need to perform mathematical and symbolic reasoning
- They do this by generating code

The risk

- This code is inherently untrusted. It could be malicious or have the ability to harm your system

The goal

- We need a way to execute this code without without compromising security

Our Quest - The Holy Grail of Secure, Ephemeral Execution

What we need ... a sandbox that is

Securely isolated
to run untrusted code safely

Fast and ephemeral
that is instantly available and
cleaned up

Scalable
that is ready for one user or
thousands



Enter:

Azure Container Apps Dynamic Sessions

The core concept – A real sandbox for your code



Dynamic sessions

- Fast, isolated execution environments.

Powered by the Hyper-V boundary:

- Hardware-enforced isolation
- Built for hostile, multi-tenant scenarios
- Keeps workloads strictly separated



Two Flavors of Sessions

1. Built-in Code Interpreter
 - A ready-to-use Python environment.
 - Comes with pre-installed packages like NumPy, pandas, and scikit-learn.
 - Perfect when you don't want to manage your own containers.
2. Custom Container Sessions
 - Bring Your Own Environment with a custom container image.
 - Use any language or dependency you need.
 - Option for additional network isolation.

AZURE RESOURCE

Container Apps Session Pool ...

Basics Tags Review + create

Execute untrusted code at scale with Azure Container Apps sessions. Sessions are ephemeral and secure containers with full Hyper-V and network isolation. To get started, create a session pool. [Learn more](#)

Project details

Select a subscription to manage resource creation and costs, and a resource group to organize all your resources for this deployment.

Subscription *	MVP Azure Sponsorship 2025-2026
Resource group *	cp-vslive-secure-code-exec Create new resource group

Session pool details

Session pool name *	cp-vs-live
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Pool type *	☒ Python code interpreter Execute code in sessions with a platform-managed Python interpreter ☐ Node.js code interpreter (preview) Execute code in sessions with a platform-managed Node.js interpreter ☐ Custom container Run sessions with your own container image
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Language type	Python
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Maximum concurrent sessions * ⓘ	5
---------------------------------	--------------

Session cooldown period * ⓘ	300 seconds
-----------------------------	-----------------------------------

Network egress ⓘ	☒ Enabled
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Region *	East US
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The Session Pool Architecture



Session Pool: The Azure resource that manages your sessions



Configuration: You define settings like max concurrent sessions and the idle "cooldown period"

How it works



1. Azure pre-warms container instances and holds them in the pool.



2. When your app requests a session, one is instantly allocated.



3. After the cooldown period of inactivity, the session is automatically stopped and destroyed

Demo

- Azure resource walk-through
- Playground

Security Model: Authentication



Every request authenticated with Microsoft Entra (Azure AD) tokens.

Audience claim must be <https://dynamicsessions.io>.

Frameworks like LangChain/Semantic Kernel handle token management automatically.

Security Model – Authorization

Requests must be authorized via Azure RBAC.



Required roles:

Azure Container Apps Session
Executor

Contributor

Contributor (1)

☐	 Carey Payette cpayette@trilliuminnovations.com	User	Contributor	 Resource group (Inherited)
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Azure ContainerApps Session Executor (2)

☐	 Carey Payette cpayette@trilliuminnovations.com	User	Azure ContainerApps Session Executor	 This resource
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True isolation: The secret handshake



The session identifier is a unique string you define to reference a session (e.g., a user ID or conversation ID).



It allows you to reuse the same session for subsequent requests.

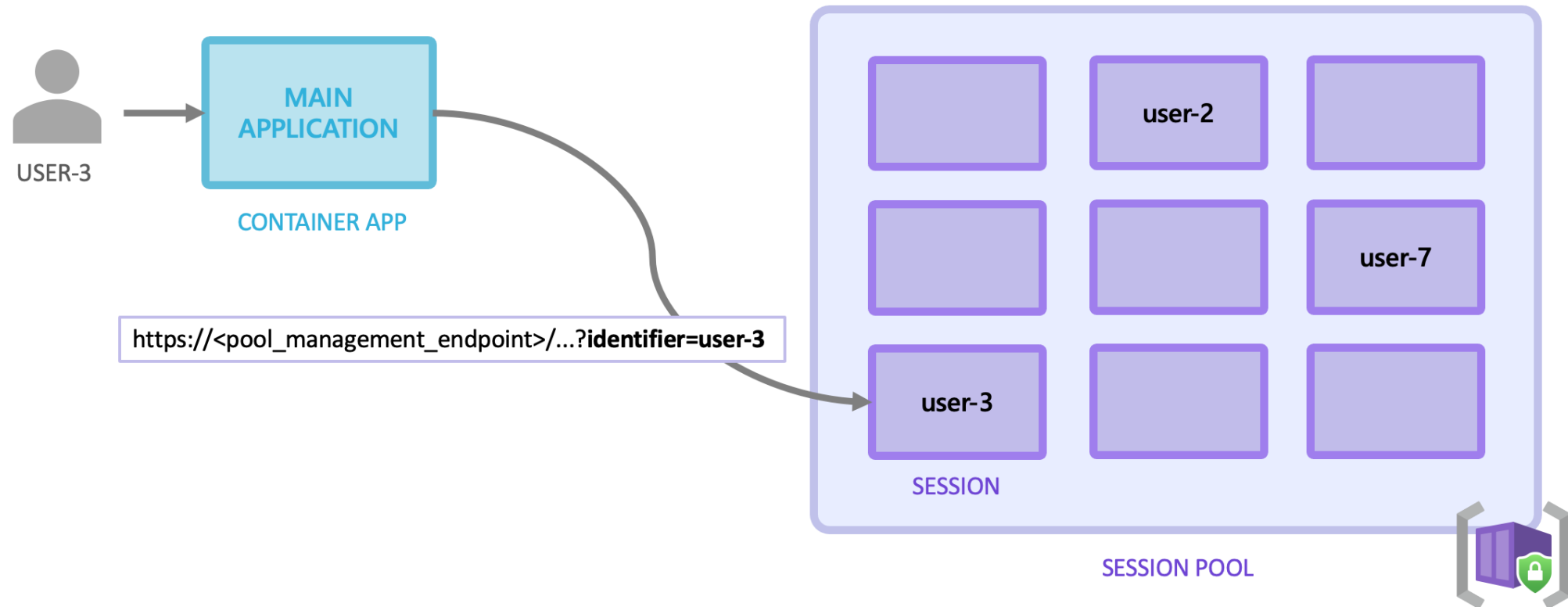


CRITICAL WARNING: The session identifier is sensitive information.



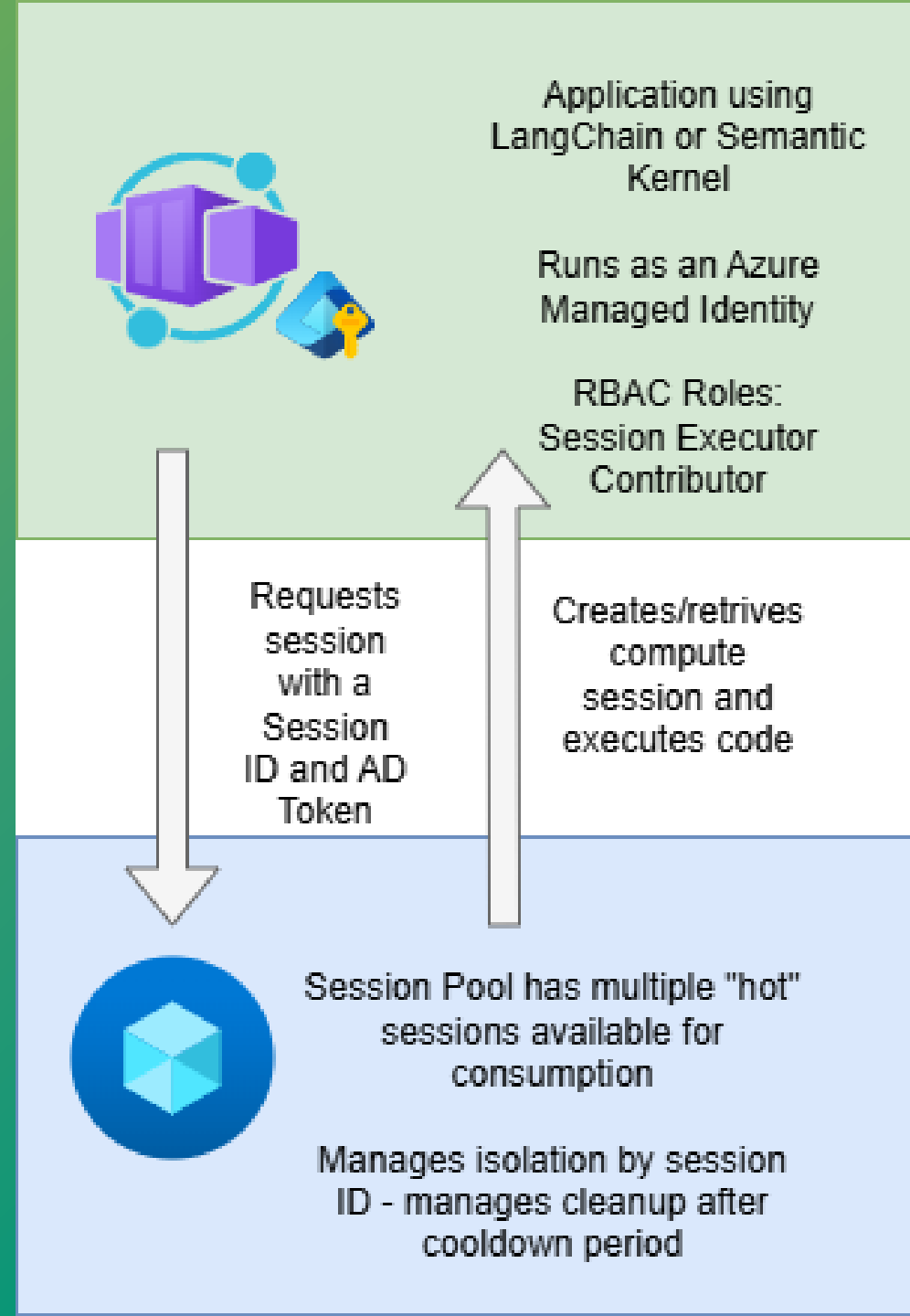
Your Responsibility: Your application must ensure a user can only access their own session identifier to prevent unauthorized data access.

Session Identifiers



SECURITY RECAP

+
○



Demos

REST

LangChain

Semantic Kernel

Microsoft Agent Framework

MCP Server in Foundry

Custom Container



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